

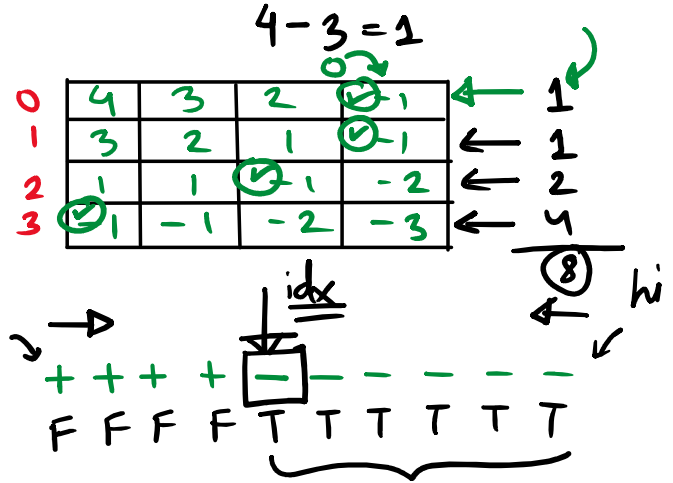
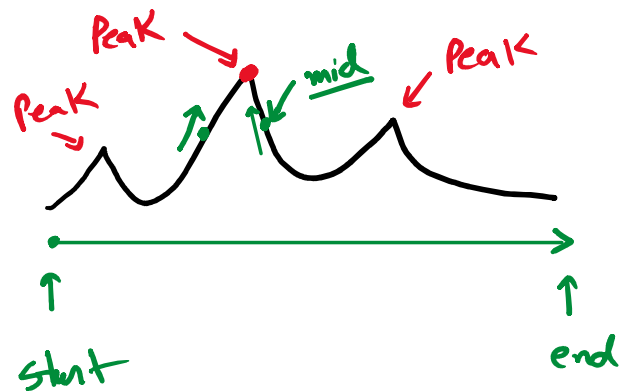
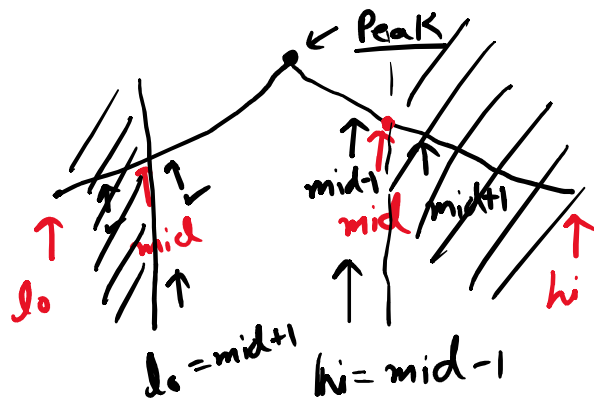
Input: grid = [[4,3,2,-1],[3,2,1,-1],[1,1,-1,-2],[-1,-1,-2,-3]]
Output: 8

From <<https://leetcode.com/problems/count-negative-numbers-in-a-sorted-matrix/>>

T.C: $O(R \cdot \log C)$

0 58 upper-bound

0 1 2 3 4 5 6 7
2, 4, 5, 7, 4, 3, 2, 1



→ Mountain Array

Condition 1: PEAK

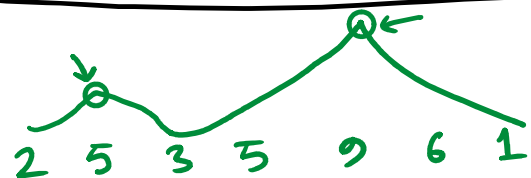
$arr[mid-1] < arr[mid]$
 $arr[mid+1] < arr[mid]$ ← Peak

Condition 2: $hi = mid - 1$

$arr[mid-1] > arr[mid]$
 $arr[mid+1] < arr[mid]$

Condition 3: $lo = mid + 1$

$arr[mid-1] < arr[mid]$
 $arr[mid+1] > arr[mid]$



$arr[mid-1] > arr[mid]$
peak is on the left

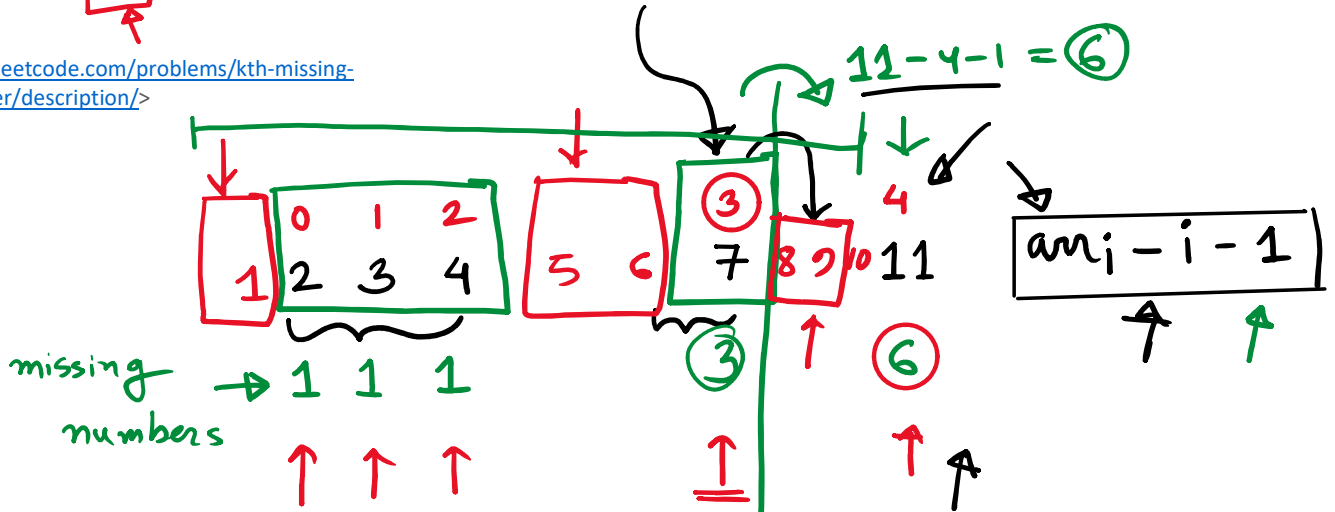
... mid

peak is on ...

$arr[mid+1] > arr[mid]$
peak is on the Right

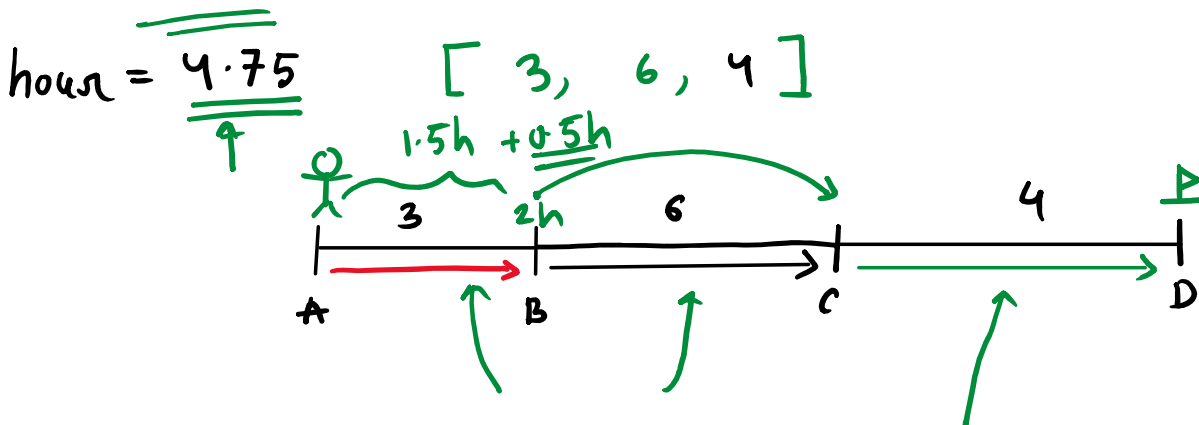
Input: $arr = [2, 3, 4, 7, 11]$, $k = 5$
Output: 9

From <https://leetcode.com/problems/kth-missing-positive-number/description/>



of values exist + # of values don't exist = the missing value
4 + 5 = 9

all values exist before 'K' + K = the K'th missing value
 $4 + 5 = 9$



$3/2 = 1.5h$ min Speed = 2

